

Comparing Penalized VAR Approaches for Intensive Longitudinal Data

STAT 293 Final Project

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Outline

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Introduction

What is Intensive Longitudinal Data (ILD)?

- ILD refers to data collected repeatedly over short intervals, they're often collected daily or multiple times per day.
- Common sources include physiological measurements, health and activity data from sensors, and assessments of behavioral and emotional states.
- ILD captures short-term dynamics in mood, behavior, health, and psychological processes.
- These datasets typically introduce several challenges:
 - Short time series per person
 - Many variables compared to available observations
 - Heterogeneity across individuals

Penalized Estimation and Forecasting of Multiple Subject Intensive Longitudinal Data by Fisher, Kim, Fredrickson, and Pipiras (2022).

- The paper proposes a multi-VAR approach for modeling several individuals at once while allowing each person to have partially unique dynamics.
- A vector autoregressive (VAR) model predicts each variable using its own past values and the past values of other variables.
- The short, high-dimensional nature of ILD increases the risk of overfitting, so penalties like LASSO help stabilize and simplify the model.

Statistical Problem

- The main problem addressed in this project is how to estimate and forecast high-dimensional ILD when each individual has a short time series and potentially different underlying dynamics.
- This motivates several key questions:
 - How can we model multiple individuals jointly while still allowing person-specific effects?
 - How do penalized VAR methods improve estimation in high dimensions?
 - Can multi-VAR approaches yield better forecasting accuracy than standard single-subject VAR models?
- Addressing these questions helps improve forecasting in psychological, behavioral, and health applications using ILD.

What We Do in This Project

- We evaluate four different penalized VAR methods for modeling ILD.
 - LASSO VAR (Lag 1)
 - LASSO VAR (Lag 2)
 - Multi-individual VAR (multi-VAR)
 - Adaptive multi-VAR
- A “lag” captures how earlier observations in the series help predict the current measurement.
- Our goal is to understand how these approaches perform in:
 - Recovering true underlying dynamics (Sensitivity, Specificity, Estimation Error)
 - Forecasting accuracy across model settings (Root Mean Squared Forecast Error)

Methodology

Vector Autoregressive (VAR) Models

Consider a multivariate time series $\{\mathbf{X}_t\}_{t \in \mathbb{Z}}$, where each $\mathbf{X}_t = (X_{1,t}, \dots, X_{d,t})'$ is a d -dimensional column vector.

VAR(p) model:

$$\mathbf{X}_t = \Phi_1 \mathbf{X}_{t-1} + \dots + \Phi_p \mathbf{X}_{t-p} + \varepsilon_t, \quad t \in \mathbb{Z}, \quad (1)$$

where $\Phi_1, \dots, \Phi_p$ are $d \times d$ coefficient matrices.

Noise assumptions:

- $\{\varepsilon_t\}$ is white noise, $\varepsilon_t \sim \text{WN}(0, \Sigma_\varepsilon)$.
- $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[\varepsilon_t \varepsilon_s'] = 0$ for $s \neq t$.

Stability and stationarity:

- Define $\Phi(z) = I_d - \Phi_1 z - \dots - \Phi_p z^p$.
- A unique causal stationary solution exists if $\det(\Phi(z)) \neq 0$ for all $|z| \leq 1$, $z \in \mathbb{C}$.

Estimation of Unrestricted VAR(p)

- VAR(p) model with sample size T :

$$\mathbf{X}_t = \Phi_1 \mathbf{X}_{t-1} + \cdots + \Phi_p \mathbf{X}_{t-p} + \varepsilon_t, \quad t = p+1, \dots, T.$$

- ordinary least squares (OLS) estimator:

$$(\hat{\Phi}_1, \dots, \hat{\Phi}_p) = \arg \min_{\Phi_1, \dots, \Phi_p} \sum_{t=p+1}^T \|\mathbf{X}_t - \Phi_1 \mathbf{X}_{t-1} - \cdots - \Phi_p \mathbf{X}_{t-p}\|_2^2.$$

- $\|\cdot\|_2$ is the Frobenius norm, so OLS is equivalent to running separate least squares regressions for each of the d component series.
- Drawback: the number of parameters $pd^2 + d(d-1)/2$ in an unrestricted VAR(p) grows quickly with d and p , which makes estimation difficult in high dimensions.

Estimation of Sparse VAR Models

Motivation. In high-dimensional VAR(p) models, the number of parameters is large, so we impose sparsity on the coefficient matrices $\Phi_1, \dots, \Phi_p$ and use penalized estimation.

Regression form. Let $N = T - p$ and stack the data as

$$Y = \begin{pmatrix} \mathbf{X}'_{p+1} \\ \mathbf{X}'_{p+2} \\ \vdots \\ \mathbf{X}'_T \end{pmatrix}, \quad Z = \begin{pmatrix} \mathbf{X}'_p & \mathbf{X}'_{p-1} & \cdots & \mathbf{X}'_1 \\ \mathbf{X}'_{p+1} & \mathbf{X}'_p & \cdots & \mathbf{X}'_2 \\ \vdots & \vdots & & \vdots \\ \mathbf{X}'_{T-1} & \mathbf{X}'_{T-2} & \cdots & \mathbf{X}'_{T-p} \end{pmatrix}, \quad B^* = \begin{pmatrix} \Phi'_1 \\ \Phi'_2 \\ \vdots \\ \Phi'_p \end{pmatrix}$$

and $E = (\epsilon'_{p+1}, \dots, \epsilon'_T)'$. Then the VAR model can be written as

$$Y = ZB^* + E, \quad \text{or} \quad \vec{Y} = (I_d \otimes Z)\vec{B^*} + \vec{E},$$

where $Y, E \in \mathbb{R}^{N \times d}$, $Z \in \mathbb{R}^{N \times pd}$, $B^* \in \mathbb{R}^{pd \times d}$.

Toy Example: VAR(2) in Regression Form

Setup. Consider a bivariate VAR(2)

$$\mathbf{X}_t = \Phi_1 \mathbf{X}_{t-1} + \Phi_2 \mathbf{X}_{t-2} + \varepsilon_t, \quad t = 3, \dots, 4,$$

where $\mathbf{X}_t = (\mathbf{X}_{1,t}, \mathbf{X}_{2,t})' \in \mathbb{R}^2$ and $\Phi_1, \Phi_2 \in \mathbb{R}^{2 \times 2}$.

Suppose we observe

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Stacked regression form. Here $T = 4$ and $p = 2$, so $N = T - p = 2$.

$$Y = \begin{pmatrix} \mathbf{x}'_3 \\ \mathbf{x}'_4 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad Z = \begin{pmatrix} \mathbf{x}'_2 & \mathbf{x}'_1 \\ \mathbf{x}'_3 & \mathbf{x}'_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}.$$

Let

$$B^* = \begin{pmatrix} \Phi'_1 \\ \Phi'_2 \end{pmatrix} \in \mathbb{R}^{4 \times 2}, \quad E = \begin{pmatrix} \varepsilon'_3 \\ \varepsilon'_4 \end{pmatrix}.$$

Then the VAR(2) model can be written as the multivariate regression

$$Y = ZB^* + E,$$

where $Y, E \in \mathbb{R}^{N \times 2}$, $Z \in \mathbb{R}^{N \times 4}$, and $B^* \in \mathbb{R}^{4 \times 2}$.

Method (a): LASSO Estimation

Assuming B is s -sparse, i.e., it has s nonzero entries.

The **LASSO estimator** can be written as

$$\hat{B} = \arg \min_B \frac{1}{N} \|Y - ZB\|_F^2 + \lambda \|B\|_1,$$

where $\|\cdot\|_F$ is the Frobenius norm and $\|B\|_1 = \sum_{j,k} |B_{jk}|$.

Larger λ yields sparser VAR coefficient matrices.

This is equivalent to running separate LASSO regressions for each of the d component series.

Method (b): The multi-VAR Approach

Decomposition

$$B_k^* = \mu^* + \Delta_k^*, \quad k = 1, \dots, K,$$

where $\mu^* \in \mathbb{R}^q$ corresponds to the common effects across K individuals and $\Delta_k^* \in \mathbb{R}^q$ corresponds to the effects unique to individual k . $q = dp^2$

The **multi-VAR LASSO estimator** can be written as

$$(\hat{\mu}, \hat{\Delta}_1, \dots, \hat{\Delta}_K) = \arg \min_{\mu, \Delta_1, \dots, \Delta_K} \left\{ \frac{1}{N} \sum_{k=1}^K \|Y^{(k)} - Z^{(k)}(\mu + \Delta_k)\|_2^2 + \lambda_1 \|\mu\|_1 + \sum_{k=1}^K \lambda_{2,k} \|\Delta_k\|_1 \right\}.$$

Method (c): Adaptive multi-VAR approach

The **multi-VAR adaptive-LASSO estimator** can be written as

$$(\hat{\mu}, \hat{\Delta}^{(1)}, \dots, \hat{\Delta}^{(K)}) = \arg \min_{\mu, \Delta^{(1)}, \dots, \Delta^{(K)}} \left\{ \frac{1}{N} \sum_{k=1}^K \|Y^{(k)} - Z^{(k)}(\mu + \Delta^{(k)})\|_2^2 + \lambda_1 \left(\omega \|\mu\|_1 + \sum_{k=1}^K \frac{\lambda_{2,k}}{\lambda_1} v_k \|\Delta^{(k)}\|_1 \right) \right\},$$

where

- μ is the population mean coefficient vector and $\Delta^{(k)}$ is the subject specific deviation for individual k .
- $\omega = (\omega_j)$ and $v_k = (v_{k,j})$ are positive weights constructed from a pilot estimate $\tilde{B}$:

$$\omega_j = \frac{1}{|\tilde{B}_{\cdot j}|}, \quad v_{k,j} = \frac{1}{|\hat{B}_{k,j} - \tilde{B}_{\cdot j}|}.$$

A nice property of multi-VAR approach is that they can be written as a weighted LASSO problem.

Implementation

Paper Implementation: Simulation Design

- Goal: evaluate forecasting performance for multiple subject ILD using penalized VAR models.
- Design factors:
 - Time series length: $T \in \{30, 50, 100\}$.
 - Number of ILD variables: $d \in \{10, 20, 30\}$.
 - Number of individuals: $K \in \{10, 20\}$.
 - Cross sectional heterogeneity: low / medium / high.
 - VAR method: (a) individual LASSO VAR, (b) multi-VAR, (c) adaptive multi-VAR.
- Data generating model:
 - Each individual has a $d \times d$ sparse transition matrix with 5% nonzero entries.
 - Nonzero locations chosen randomly by heterogeneity level.
 - Low: 2/3 of paths shared, 1/3 unique.
 - Medium: 1/2 shared, 1/2 unique.
 - High: 1/3 shared, 2/3 unique.
 - Nonzero coefficients i.i.d. from $U(0.1, 0.9)$ until the VAR stability condition holds.

Evaluation:

- Last 3 time points of each series held out for forecast accuracy.
- For synthetic data, cross validation is done in two ways:
 - Using the true transition matrix to choose tuning parameters that minimize estimation error.
 - Using the rolling-window cross-validation (RWCV) procedure to select λ_1 and λ_2 under realistic conditions.
- Outcome Measures:
 - **Estimation Error** – how close the estimated coefficient matrices are to the true underlying VAR coefficients.
 - **Sensitivity** – ability to correctly detect true nonzero effects.
 - **Specificity** – ability to correctly identify zero effects.
 - **RMSFE** – out-of-sample forecast error.

Paper Results

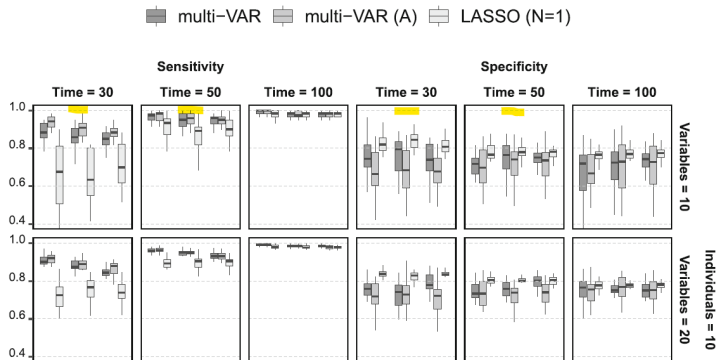


Fig2. Sensitivity and specificity for B_k (rolling window cross-validation).

Our Implementation: Simulation Settings Overview

We also run simulations with $p = 2$, which is not considered in the original paper.

Settings with $p = 1$

Setting	K	T	d	p
1	10	30	10	1
3	10	30	20	1
5	10	50	10	1
7	10	50	20	1
9	20	30	10	1
11	20	30	20	1
13	20	50	10	1
15	20	50	20	1

Settings with $p = 2$

Setting	K	T	d	p
2	10	30	10	2
4	10	30	20	2
6	10	50	10	2
8	10	50	20	2
10	20	30	10	2
12	20	30	20	2
14	20	50	10	2
16	20	50	20	2

Notation: K number of individuals
 T number of time points
 d number of variables
 p VAR lag order

Common Parameter Values for Simulation Settings

Common Parameter Values Used in All Simulation Settings

Factor	Value	Description
sparsity	0.05	Proportion of non-zero coefficients
heterogeneity	medium	Level of heterogeneity (low/medium/high)
sigma_eps	0.1	Standard deviation of noise
phi_max	0.9	Upper bound for coefficient generation
phi_min	0.1	Lower bound for coefficient generation
n_holdout	5	Number of time points withheld for forecasting

Simulation Setup: Data Generation and Estimation

- **Burn in strategy.** For each subject we first generate a burn in period of $T_{\text{burn}} = 50$ time points and discard them, then keep the remaining T observations for analysis.
- **Stability condition and scaling.** We require the VAR process to satisfy the stability condition (spectral radius of the companion matrix < 1). If after 100 random draws the process is still unstable, we rescale: let $\lambda_{\max}$ be the largest eigenvalue and set

$$c = \frac{\lambda_{\max}}{0.95}, \quad \Phi_{\text{new}} = \Phi_{\text{old}}/c.$$

This forces the spectral radius to be below the target 0.95.

- **Estimation methods:**
 - LASSO: `glmnet` package.
 - multi-VAR: `multivar` package.

Implementation Details

- For individual LASSO, we stack the VAR data into regression form and apply `cv.glmnet`.
- Code Arguments for Applying the Multi-VAR Approach :
 - `lambda1`: matrix of penalty parameters λ_1 (default: NULL).
 - `lambda2`: matrix of penalty parameters λ_2 (default: NULL).
 - `nlambda1`: number of λ_1 values in the grid (default: 30).
 - `nlambda2`: number of λ_2 values in the grid (default: 30).
 - `cv`: cross-validation type; "rolling" (default) or "blocked".
 - `window`: rolling-window size for cross-validation.
 - `nfolds`: number of folds when `cv` = "blocked".
- We use blocked cross-validation (`cv` = "blocked", `nfolds` = 10) to tune the penalty parameters, whereas the original paper uses RWCV (`cv` = "rolling").
- A main challenge in our simulation study is selecting suitable values for λ_1 and λ_2 .

Data Analysis

Our Implementation: Simulation Settings Overview

We also run simulations with $p = 2$, which is not considered in the original paper.

Settings with $p = 1$

Setting	K	T	d	p
1	10	30	10	1
3	10	30	20	1
5	10	50	10	1
7	10	50	20	1
9	20	30	10	1
11	20	30	20	1
13	20	50	10	1
15	20	50	20	1

Settings with $p = 2$

Setting	K	T	d	p
2	10	30	10	2
4	10	30	20	2
6	10	50	10	2
8	10	50	20	2
10	20	30	10	2
12	20	30	20	2
14	20	50	10	2
16	20	50	20	2

Notation: K number of individuals
 T number of time points
 d number of variables
 p VAR lag order

Simulation Results: $P = 1$

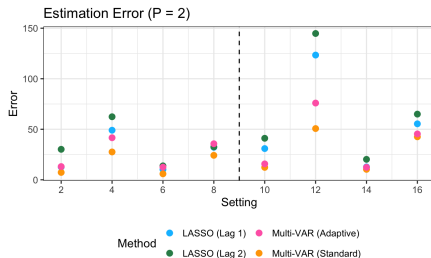
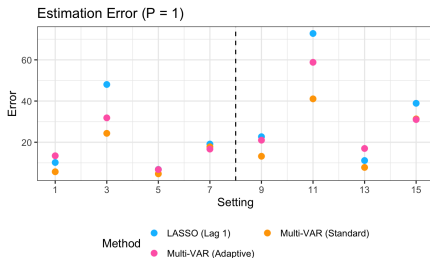
Setting	Method	Est. Error	Sensitivity	Specificity	RMSFE
1	LASSO(1)	10.18	0.72 (0.193)	0.837 (0.057)	0.104 (0.015)
	Multi-VAR Std	5.68	0.86 (0.097)	0.818 (0.027)	0.105 (0.016)
	Multi-VAR Adap	13.43	0.14 (0.232)	0.998 (0.007)	0.101 (0.013)
3	LASSO(1)	48.05	0.705 (0.128)	0.855 (0.053)	0.113 (0.014)
	Multi-VAR Std	24.33	0.905 (0.050)	0.800 (0.012)	0.109 (0.012)
	Multi-VAR Adap	31.84	0.52 (1.757)	0.978 (0.029)	0.107 (0.012)
5	LASSO(1)	6.82	0.90 (0.141)	0.827 (0.053)	0.107 (0.009)
	Multi-VAR Std	4.67	0.90 (0.105)	0.719 (0.039)	0.104 (0.008)
	Multi-VAR Adap	6.75	0.44 (0.280)	0.995 (0.010)	0.105 (0.009)
7	LASSO(1)	19.12	0.840 (0.074)	0.855 (0.045)	0.104 (0.009)
	Multi-VAR Std	17.79	0.940 (0.032)	0.777 (0.045)	0.104 (0.008)
	Multi-VAR Adap	16.69	0.675 (0.086)	0.985 (0.008)	0.104 (0.007)
9	LASSO(1)	22.68	0.710 (0.117)	0.857 (0.049)	0.104 (0.013)
	Multi-VAR Std	13.20	0.820 (0.114)	0.833 (0.026)	0.109 (0.016)
	Multi-VAR Adap	21.00	0.070 (0.198)	0.997 (0.010)	0.102 (0.011)
11	LASSO(1)	72.79	0.645 (0.131)	0.868 (0.045)	0.112 (0.011)
	Multi-VAR Std	41.07	0.850 (0.069)	0.795 (0.011)	0.112 (0.015)
	Multi-VAR Adap	58.81	0.460 (0.137)	0.986 (0.021)	0.106 (0.009)
13	LASSO(1)	11.12	0.610 (0.229)	0.844 (0.072)	0.101 (0.008)
	Multi-VAR Std	7.78	0.930 (0.098)	0.816 (0.037)	0.101 (0.007)
	Multi-VAR Adap	16.98	0.070 (0.163)	0.999 (0.002)	0.102 (0.007)
15	LASSO(1)	38.91	0.753 (0.092)	0.857 (0.036)	0.109 (0.008)
	Multi-VAR Std	31.33	0.857 (0.036)	0.790 (0.016)	0.107 (0.007)
	Multi-VAR Adap	31.01	0.573 (0.106)	0.991 (0.009)	0.107 (0.007)

Simulation Results: $P = 2$

Setting	Method	Est. Error	Sensitivity	Specificity	RMSFE
2	LASSO(1)	12.65	0.78 (0.175)	0.849 (0.058)	0.110 (0.013)
	LASSO(p)	30.17	0.53 (0.125)	0.838 (0.055)	0.114 (0.017)
	Multi-VAR Std	7.30	0.84 (0.184)	0.694 (0.013)	0.110 (0.014)
	Multi-VAR Adap	13.09	0.24 (0.350)	0.995 (0.009)	0.108 (0.012)
4	LASSO(1)	49.053	0.69 (0.094)	0.817 (0.071)	0.210 (0.074)
	LASSO(p)	62.377	0.693 (0.049)	0.869 (0.032)	0.195 (0.079)
	Multi-VAR Std	27.552	0.875 (0.042)	0.614 (0.093)	0.815 (1.293)
	Multi-VAR Adap	41.618	0.65 (0.082)	0.946 (0.040)	0.722 (1.075)
6	LASSO(1)	10.229	0.72 (0.270)	0.799 (0.084)	0.109 (0.012)
	LASSO(p)	13.787	0.69 (0.213)	0.854 (0.037)	0.104 (0.016)
	Multi-VAR Std	5.904	0.90 (0.141)	0.858 (0.027)	0.108 (0.011)
	Multi-VAR Adap	12.531	0.22 (0.305)	1.000 (0.000)	0.106 (0.010)
8	LASSO(1)	32.172	0.795 (0.080)	0.789 (0.069)	0.155 (0.034)
	LASSO(p)	32.602	0.838 (0.087)	0.858 (0.018)	0.139 (0.043)
	Multi-VAR Std	24.210	0.895 (0.064)	0.562 (0.108)	1.125 (2.258)
	Multi-VAR Adap	35.669	0.680 (0.114)	0.937 (0.036)	0.991 (2.063)
10	LASSO(1)	30.903	0.680 (0.199)	0.808 (0.061)	0.117 (0.015)
	LASSO(p)	41.080	0.690 (0.117)	0.846 (0.060)	0.113 (0.015)
	Multi-VAR Std	12.288	0.800 (0.172)	0.744 (0.028)	0.116 (0.015)
	Multi-VAR Adap	15.739	0.410 (0.238)	0.982 (0.023)	0.113 (0.014)
12	LASSO(1)	123.418	0.605 (0.093)	0.830 (0.050)	0.154 (0.042)
	LASSO(p)	144.775	0.585 (0.099)	0.874 (0.023)	0.141 (0.027)
	Multi-VAR Std	50.700	0.845 (0.083)	0.785 (0.021)	0.146 (0.038)
	Multi-VAR Adap	75.942	0.380 (0.185)	0.985 (0.020)	0.159 (0.047)
14	LASSO(1)	12.713	0.870 (0.134)	0.844 (0.068)	0.120 (0.014)
	LASSO(p)	20.240	0.895 (0.089)	0.858 (0.051)	0.111 (0.010)
	Multi-VAR Std	10.356	0.900 (0.121)	0.791 (0.037)	0.121 (0.015)
	Multi-VAR Adap	12.331	0.630 (0.236)	0.997 (0.008)	0.121 (0.015)
16	LASSO(1)	55.389	0.833 (0.104)	0.808 (0.054)	0.135 (0.026)
	LASSO(p)	64.988	0.790 (0.086)	0.868 (0.024)	0.117 (0.016)
	Multi-VAR Std	42.537	0.942 (0.063)	0.581 (0.024)	0.243 (0.495)
	Multi-VAR Adap	45.384	0.750 (0.105)	0.959 (0.025)	0.270 (0.598)

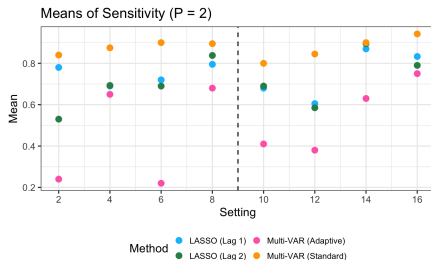
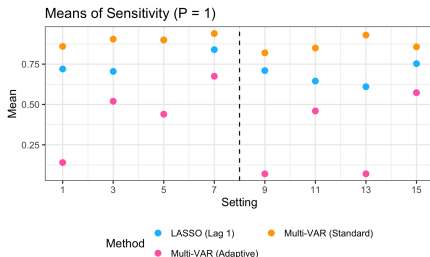
Simulation Results: Estimation Error

- Multi-VAR (Standard) consistently gets the lowest estimation error.
- LASSO methods have substantially higher estimation error, especially as K size increases.
- Adaptive Multi-VAR performs better than LASSO but it's still higher results than Standard Multi-VAR.
- Overall, Multi-VAR (Standard) has the smallest estimation error across settings.



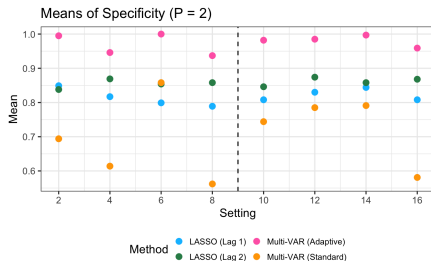
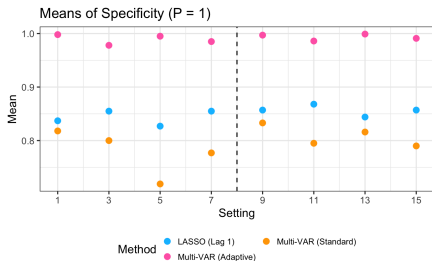
Simulation Results: Sensitivity

- Multi-VAR (Standard) has the highest sensitivity across all settings
- LASSO ($p=1$) performs reasonably well, but drops as K increases.
- LASSO ($p=2$) performs similarly to LASSO ($p=1$), except for simplest setting.
- Adaptive Multi-VAR is highly unstable for many settings, with very low results in several cases.
- Overall, the Standard Multi-VAR model is the most reliable at detecting true nonzero effects.



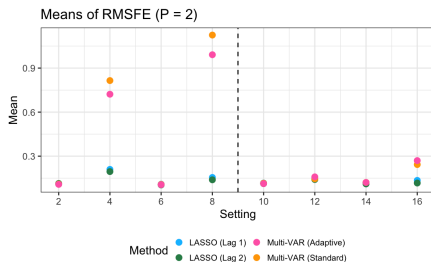
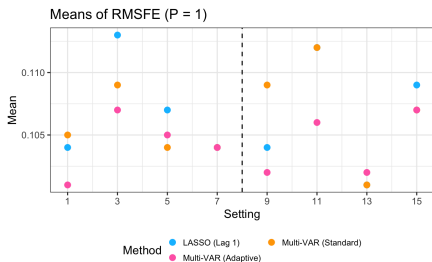
Simulation Results: Specificity

- Adaptive Multi-VAR shows the highest specificity across all settings, often above 0.98, meaning it almost never produces false positives.
- LASSO models ($p=1,2$) achieve strong and stable specificity throughout all settings
- Standard Multi-VAR has noticeably lower specificity, especially when $p = 2$, where its performance drops substantially in several settings.
- Overall, Adaptive Multi-VAR is the best at avoiding false positives, while Standard Multi-VAR trades specificity for higher sensitivity.



Simulation Results: RMSFE

- For $p = 1$, all methods perform similarly with RMSFE around 0.10.
- For $p = 2$, Multi-VAR models become unstable in several settings.
- LASSO remains consistently stable across all settings.
- Overall: similar performance for $p = 1$, but LASSO is more reliable when model complexity increases.



Conclusion

Model Summary

- **Multi-VAR (Standard)** – lowest estimation error and highest sensitivity; strongest overall for detecting true effects.
- **Adaptive Multi-VAR** – highest specificity; best for avoiding false positives.
- **LASSO** – most stable forecasting (RMSFE), especially for higher lag models ($p = 2$).

Key Questions: Findings

- **How can we model multiple individuals jointly while still allowing person-specific effects?** Multi-VAR (Standard) had the lowest estimation error and highest sensitivity across most settings, showing it effectively models all individuals together while preserving person-specific dynamics.
- **How do penalized VAR methods improve estimation in high dimensions?** LASSO methods maintained stable performance even as parameters increased, shrinking unnecessary coefficients and avoiding overfitting
- **Do multi-VAR approaches improve forecasting compared to single-subject VARs?** No. RMSFE results show all methods perform similarly for $p = 1$, but for $p = 2$, LASSO was the most stable, producing the lowest and most reliable out-of-sample forecast errors.

- Test larger parameters (individuals, variables, time points, lags) in simulations to assess scalability and robustness in high-dimensional settings.
- Investigate methods to better balance sensitivity and specificity, especially for Adaptive Multi-VAR.
- Apply multi-VAR and penalized VAR methods to real intensive longitudinal datasets, e.g., physiological or behavioral sensor data.

Thank you!

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